

Data Dashboard Mirage Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	data-dashboard-mirage

Big idea: Dashboards do not simply show data; they prioritize, compress, and narrate it.

Student-Facing Prompt

Inspect Data Dashboard Mirage Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Data Dashboard Mirage Stimulus A	Student-facing stimulus 1: Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Data Dashboard Mirage Stimulus B	Student-facing stimulus 2: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Data Dashboard Mirage Reveal / Twist	Student-facing stimulus 3: Dashboard A ranks classes by average score; Dashboard B ranks them by improvement. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Dashboards do not simply show data; they prioritize, compress, and narrate it.

Actual Lesson Questions

#	Question for students
1	After inspecting Data Dashboard Mirage Stimulus A, what is your first knowledge claim?
2	Which exact detail from Data Dashboard Mirage Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	When does data visualization become argument?

Ready-to-Use Examples

- Dashboard A highlights falling late arrivals in green; Dashboard B highlights rising absences in amber using the same school-week dataset.
- Dashboard A ranks classes by average score; Dashboard B ranks them by improvement.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Activity Steps

- 1. Show students two dashboards with the same underlying data but different layouts.
- 2. Groups identify the story each dashboard seems to tell.
- 3. Students trace which design choices create that impression.
- 4. Groups redesign one dashboard for greater transparency.
- 5. Debrief how interfaces shape what users think they know.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When does data visualization become argument?
- Can a dashboard be accurate and still misleading?
- What should designers owe users who rely on visualized data?

Knowledge Questions

- When does data visualization become argument?
- How do interfaces shape knowledge?
- What makes a technological representation responsible?

Transfer Example

For example, run this activity with a dashboard with clean metrics but missing definitions and context. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...